

Energy Module: Model Design

Hyuntae Choi, Seongeun Kim, Jiheun Ha, Jin Lee,
Medina Mukhamedina, Jiyong Eom, Son H. Kim, Haewon McJeon

19th March, 2026

KAIST IAM Group

Contents

1. Introduction
2. Framework & Data
3. Economy
4. Energy System
5. Technology Choice
6. Results & Next Steps

Introduction

Motivation: What GHIM Brings

	Before	GHIM
Economy	More energy detail, less macro feedback	Energy ↔ production feedback (KLEM)
Climate	Soft coupling or one-way feedback	Two-way coupling with multiple channels
Technology	Mostly fixed cost assumptions	Two-factor learning: technology-specific RND + LBD
Resolution	~32 regions, 5-year timesteps	100+ countries, annual resolution
Solution	Solve one period at a time (no foresight)	Adjustable rolling foresight
Reporting	Post-hoc translation for IPCC	Direct AR6 IAMC-format output

Design Goals

1 Replicates SSP Baselines

Reproduces SSP trajectories before layering policy.

2 Coupled Energy-Economy w/ Foresight

KLEM production with native energy accounting.

3 Solve Less, Simulate More

Logit-based design for robust convergence.

4 IAMC-Native Reporting

Model structure mirrors IAMC reporting.

5 Transparent & Reproducible

Open-source, every parameter traceable to raw data.

6 User-Configurable

Easy to adjust resolution and nesting w/ data provided.

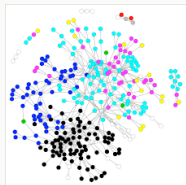
7 Open Economy

Global market clearing for key commodities.

8 Module Linkage

Defined interfaces for other GHIM modules.

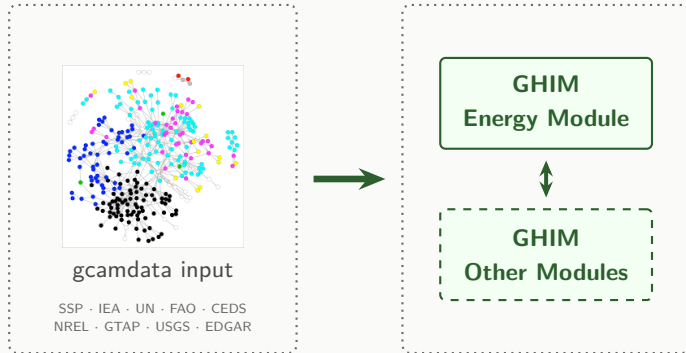
Model Workflow



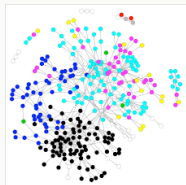
gcamdata input

SSP · IEA · UN · FAO · CEDS
NREL · GTAP · USGS · EDGAR

Model Workflow



Model Workflow



gcamdata input

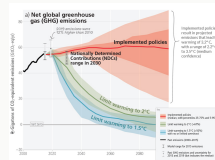
SSP · IEA · UN · FAO · CEDS
NREL · GTAP · USGS · EDGAR



GHIM
Energy Module



GHIM
Other Modules



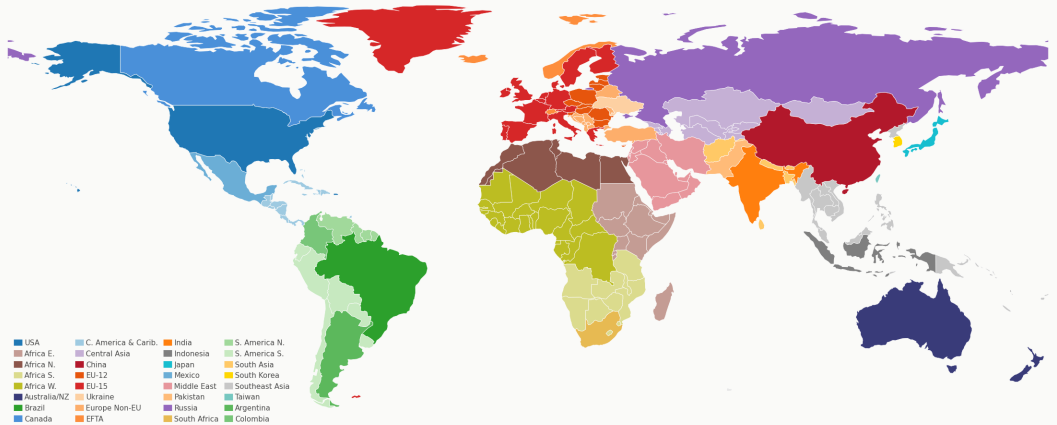
IAMC data template

Framework & Data

General Model Structure by Phase

Component	Phase 1	Phase 2+
Time	2000–2150, 5-year steps	Annual resolution
Region	32 Regions	100+ countries
Economy	$GDP = f(\text{Capital, Labor, Energy, Materials})$	+ Energy intensity $EI = f(\text{RND})$
Solution	Recursive-dynamic	Adjustable rolling foresight
Energy System	Minimum nesting for AR6 Tier-1	Expanded nesting for Tier-3
Trade	Global fossil fuel market clearing	Multi-commodity global trade
Coupling	One-way coupling with Land	Two-way with multiple modules
Tech. Choice	Absolute cost logit with SSP-calibrated preference weights	
Tech. Change	Two-factor learning: RND knowledge + cumulative deployment (LBD)	

32 Regions of Phase 1



Model Overview

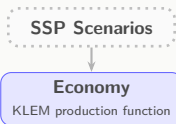
SSP Scenarios

Riahi et al. 2017

Model Overview

Riahi et al. 2017

Jorgenson & Griliches 1967



Model Overview

Riahi et al. 2017

Jorgenson & Griliches 1967

Clarke & Edmonds 1993



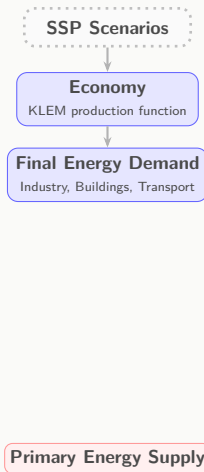
Model Overview

Riahi et al. 2017

Jorgenson & Griliches 1967

Clarke & Edmonds 1993

Rogner 1997



Model Overview

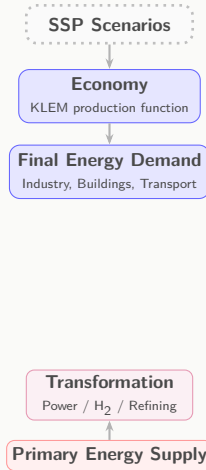
Riahi et al. 2017

Jorgenson & Griliches 1967

Clarke & Edmonds 1993

Rogner 1997

Calvin et al. 2019



Model Overview

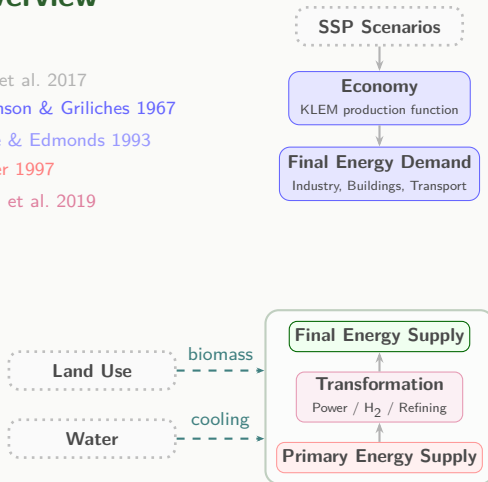
Riahi et al. 2017

Jorgenson & Griliches 1967

Clarke & Edmonds 1993

Rogner 1997

Calvin et al. 2019



Model Overview

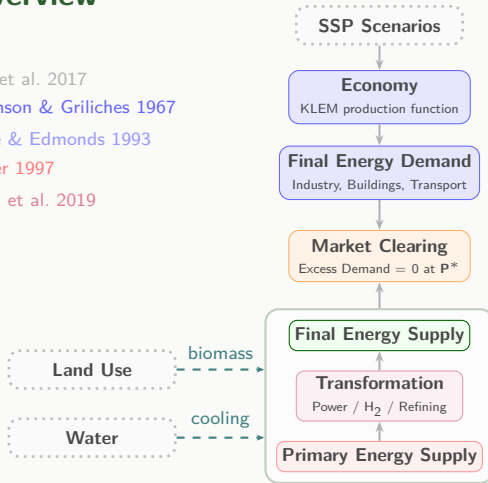
Riahi et al. 2017

Jorgenson & Griliches 1967

Clarke & Edmonds 1993

Rogner 1997

Calvin et al. 2019



Model Overview

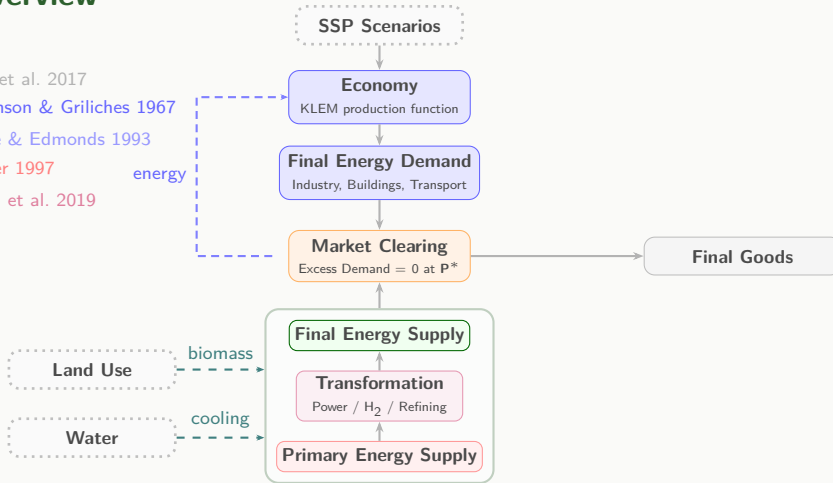
Riahi et al. 2017

Jorgenson & Griliches 1967

Clarke & Edmonds 1993

Rogner 1997

Calvin et al. 2019



Model Overview

Riahi et al. 2017

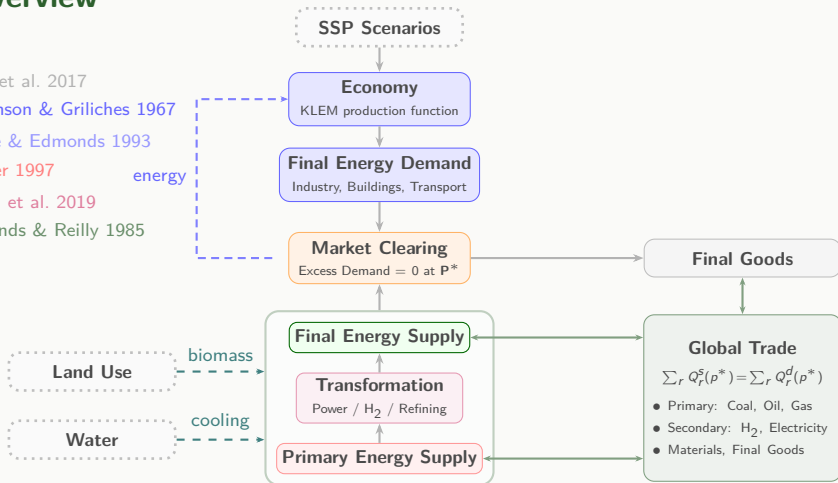
Jorgenson & Griliches 1967

Clarke & Edmonds 1993

Rogner 1997

Calvin et al. 2019

Edmonds & Reilly 1985



Model Overview

Riahi et al. 2017

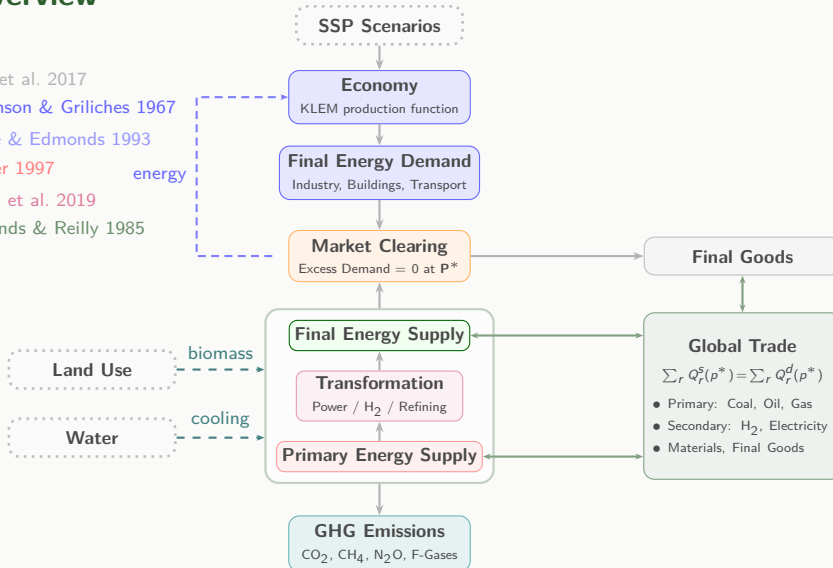
Jorgenson & Griliches 1967

Clarke & Edmonds 1993

Rogner 1997

Calvin et al. 2019

Edmonds & Reilly 1985



Model Overview

Riahi et al. 2017

Jorgenson & Griliches 1967

Clarke & Edmonds 1993

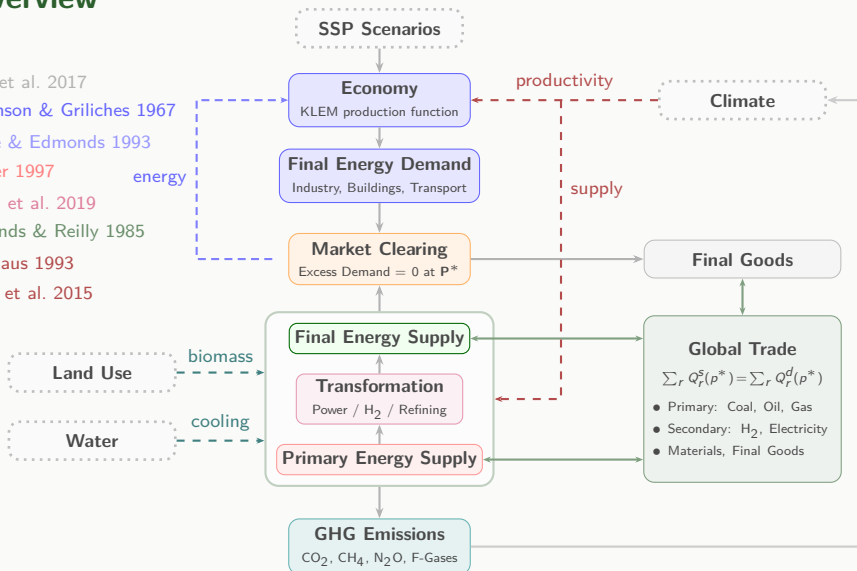
Rogner 1997

Calvin et al. 2019

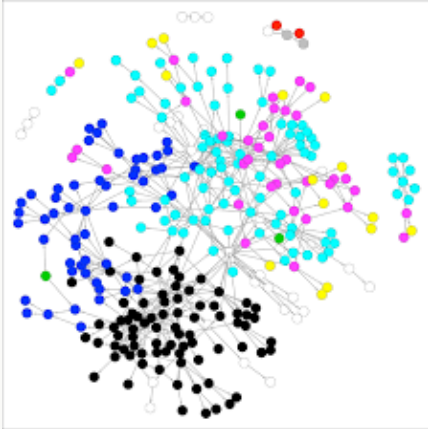
Edmonds & Reilly 1985

Nordhaus 1993

Burke et al. 2015



Input Data: The GCAM Data System

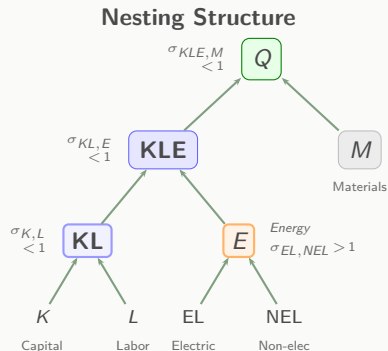


Category	Source	Version
Energy balances	IEA WEB	2023
Economy	GMD	2025
Emissions	CEDS	2022
Trade	GTAP	v11
Technology	NREL ATB	2024
Transport	UCD	2013
Land / Agriculture	FAO	2024

- IEA: International Energy Agency
- GMD: Global Macro Database
- CEDS: Community Emissions Data System
- GTAP: Global Trade Analysis Project
- NREL: National Renewable Energy Lab
- UCD: UC Davis Transportation Database
- FAO: Food & Agriculture Organization

Economy

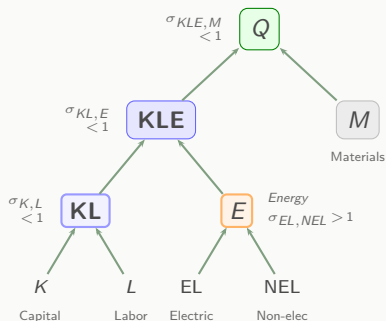
KLEM Economy (Jorgenson & Griliches, 1967; Timmer et al., 2007)



- CES: Constant Elasticity of Substitution
- Each nest has its own $\sigma \rightarrow$ heterogeneous substitution
- $\sigma < 1$: complements
- $\sigma > 1$: substitutes

KLEM Economy (Jorgenson & Griliches, 1967; Timmer et al., 2007)

Nesting Structure



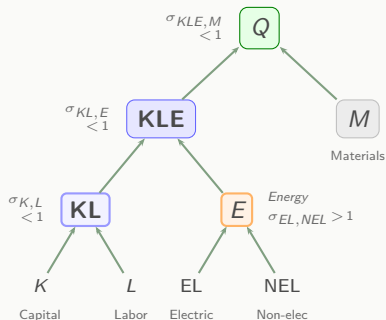
Why Nested CES?

Each nest has its own σ — heterogeneous substitution.
Carbon tax $\rightarrow$ elec replaces fossil first ($\sigma > 1$, easy),
 $\rightarrow$ then total energy falls ($\sigma < 1$, hard).

- CES: Constant Elasticity of Substitution
- Each nest has its own $\sigma \rightarrow$ heterogeneous substitution
- $\sigma < 1$: complements
- $\sigma > 1$: substitutes

KLEM Economy (Jorgenson & Griliches, 1967; Timmer et al., 2007)

Nesting Structure



- CES: Constant Elasticity of Substitution
- Each nest has its own $\sigma \rightarrow$ heterogeneous substitution
- $\sigma < 1$: complements
- $\sigma > 1$: substitutes

Why Nested CES?

Each nest has its own σ — heterogeneous substitution.
Carbon tax $\rightarrow$ elec replaces fossil first ($\sigma > 1$, easy),
 $\rightarrow$ then total energy falls ($\sigma < 1$, hard).

Key Equations

Gross Output: $Q = \text{CES}(KLE, M)$

GDP: $Y = Q - p_E \cdot E - p_M \cdot M$

Energy: E scales with output, falls when p_E rises

Budget: savings = capital + energy + R&D investment

Feedback: $p_E \uparrow \Rightarrow E \downarrow \Rightarrow Q \downarrow \Rightarrow Y \downarrow$

KLEM — Key Parameter Interpretation

$$Q = \text{CES}\left(\underbrace{\text{CES}\left(A_t \cdot \underbrace{\text{CES}(K, L; \sigma_{KL})}_{\text{KL}}, \underbrace{\text{CES}(EL, NEL; \sigma_{EL, NEL})}_{\text{E}}; \sigma_{KLE}\right)}_{\text{KLE}}, M; \sigma_Q\right)$$

Parameter	Value	Nest	Intuition
σ_{KL}	~ 0.8	KL	Complements (< 1): capital–labor (Oberfield & Raval)
$\sigma_{KL,E}$	~ 0.4	KLE	Complements (< 1): energy is essential
$\sigma_{EL,NEL}$	~ 2.0	E	Substitutes (> 1): electric replaces fossil
$\sigma_{KLE,M}$	~ 0.5	Q	Complements (< 1): materials essential
A_t	calibrated	KL	Total factor productivity; matches SSP GDP
s	0.22	Budget	GDP reinvestment rate
δ	0.05	Capital	Annual depreciation rate
ℓ	regional	Labor	Working-age share of population (ILO 2020)

Calibration — Matching SSP Trajectories

Step 1 — Input Weights α (base year, time-invariant)

Base-year spending fractions on K, L, E, M determine CES weights

Step 2 — Scale Factors A (base year, time-invariant)

Adjust each nest so CES output matches observed base-year values

Step 3 — Productivity $TFP(t)$ (solved at every future year)

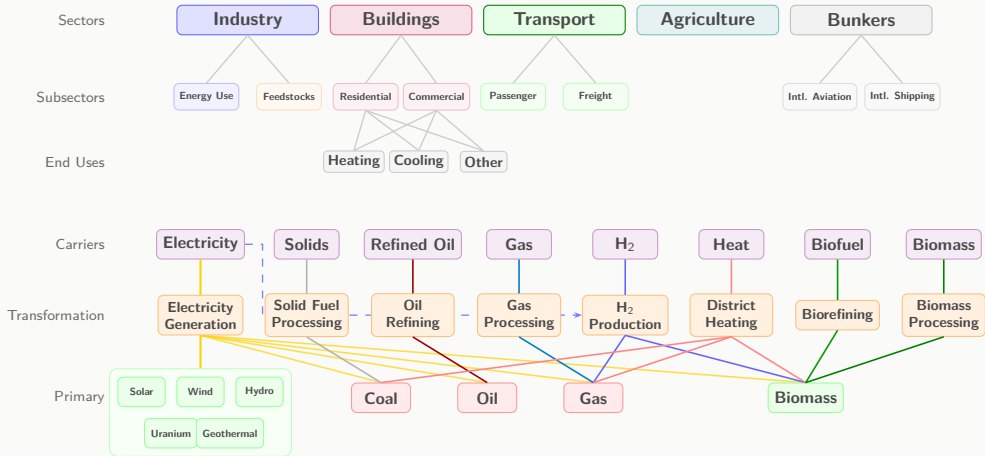
Find $TFP(t)$ so that: $Q(t) - p_E \cdot E - p_M \cdot M = GDP_{SSP}(t)$

Result

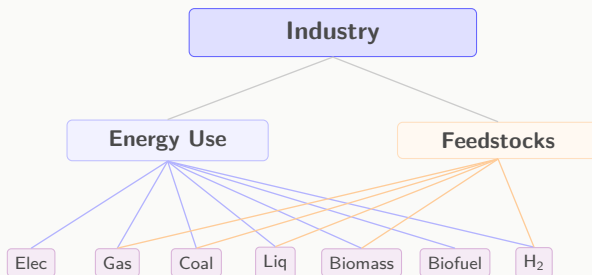
Reference run reproduces SSP GDP within tolerance ($< 0.2\%$ deviation).

Energy System

Energy System Overview (Phase 1)

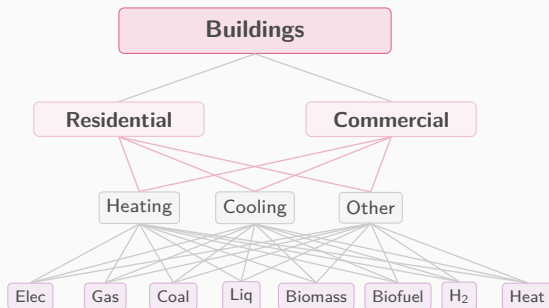


Industry



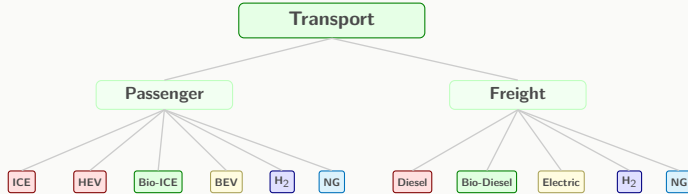
- **Demand:** $\text{Prod} = A_{\text{prod}} \cdot \text{Prod}_0 \cdot (Y/Y_0)^\alpha \cdot (P/P_0)^\beta$
- **Energy Use:** $EU = ei_{EU} \cdot \text{Prod}$
- **Feedstocks:** $FS = ei_{FS} \cdot \text{Prod}$
- **Price Index:** $P = ei_{EU} \cdot P_{EU} + ei_{FS} \cdot P_{FS}$

Buildings



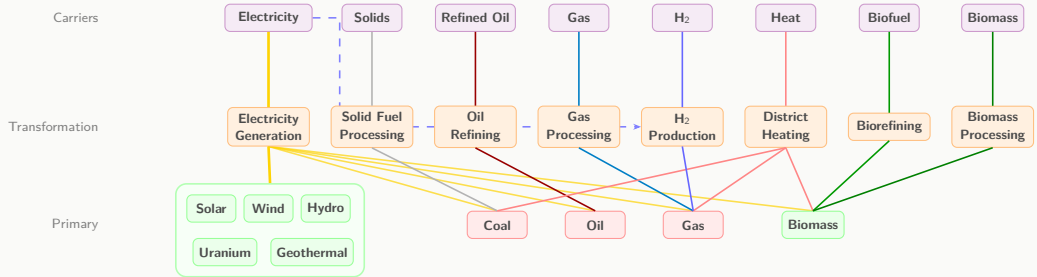
- Income $\uparrow \rightarrow$ floorspace per capita $\uparrow$ (saturates in rich countries)
- Floorspace $\times$ climate (HDD/CDD) $\rightarrow$ service demand per m²
- Service demand $\div$ carrier efficiency $\rightarrow$ final energy by carrier

Transport



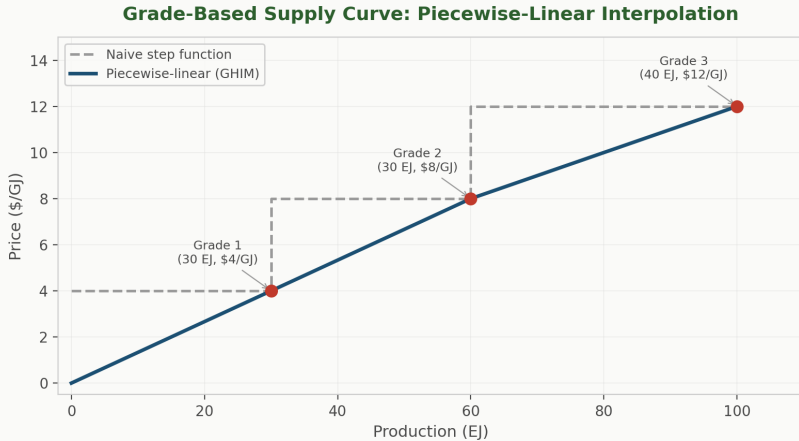
- Income $\uparrow \rightarrow$ service demand per capita $\uparrow$ (income elasticity α)
- Service demand $\div$ vehicle efficiency $\rightarrow$ final energy by carrier
- **LCOT** = LCOE + time cost: vehicle cost + fuel + O&M + $\frac{VOT}{speed}$
 - Consumers value travel time — faster/costlier modes compete with slower/cheaper ones

Energy Supply Chain



Primary Energy Supply — Fossil Fuels (Rogner, 1997)

Grade-based supply curves for coal, oil, gas per region — higher prices unlock higher-cost grades.



Inter-Regional Trade — Global Market Clearing (Edmonds & Reilly, 1985)

Global Market Clearing

- Single world price p^* per fuel (coal, oil, gas)
- $\sum_r Q_r^s(p^*) = \sum_r Q_r^d(p^*)$ (supply = demand)
- Net trade: $X_r = Q_r^s - Q_r^d$

Calibration Rent (region-specific, fixed at base year)

- $\text{rent}_r = P_{\text{observed}} - P_{\text{cleared}}$ per region r
- Carried forward as price adder into all future clearing
- Supply evaluated at extraction cost + rent_r

Technology Choice

Technology Choice (Modified from Clarke & Edmonds, 1993)

$$s_i = \frac{\alpha_i \cdot e^{\beta C_i + p_i}}{\sum_j \alpha_j \cdot e^{\beta C_j + p_j}}$$

α_i Availability Switch

Gate $\{0, 1\}$: ban / enable technologies via policy

p_i Preference Weight

SSP-calibrated non-cost factor; decays over time

C_i Cost (\$/GJ)

- LCOE: capex + O&M + fuel
- LCOT: LCOE + time value

Two-Factor Learning (Bosetti et al. 2006)

Capital cost declines with cumulative deployment and R&D:

$$\text{capex}_j(t) = \text{capex}_{j,0} \times \underbrace{\left(\frac{Q_j^{\text{cum}}}{Q_{j,0}} \right)^{-\alpha}}_{\text{LBD}} \times \underbrace{\left(\frac{H_j^{\text{RD}}}{H_{j,0}} \right)^{-\beta}}_{\text{RND}} \quad \text{Floor: } 0.2 \times \text{capex}_0$$

- **LBD** (Learning by Doing): cost ↓ with cumulative deployment Q_j^{cum}
- **RND** (Research & Development): cost ↓ with RND knowledge stock H_j^{RD}

Source: Rubin et al. 2015, *Energy Policy*; Way et al. 2022, *Joule*

Technology	LBD Rate	RND
Solar PV	20%	High
Wind	12%	Medium
Batteries	19%	High

Vintage Retirement (Grubler, 1990)

Capacity at vintage v survives only if the plant is both young enough **AND** profitable enough.

$$Q_{\text{eff}}(t) = \sum_v C_{i,v} \cdot S_i(t - v) \cdot P(\pi_{i,v})$$

Age-Based Survival $S(\text{age})$

$$S(a) = \frac{1/(1 + e^{k(a-\rho L)})}{S(0)}$$

- $\rho = 0.75$ (half-life ratio), $k = 0.1$ (steepness)
- Lifetime L differs by technology
- Renewables: hard cutoff at L (no S-curve)

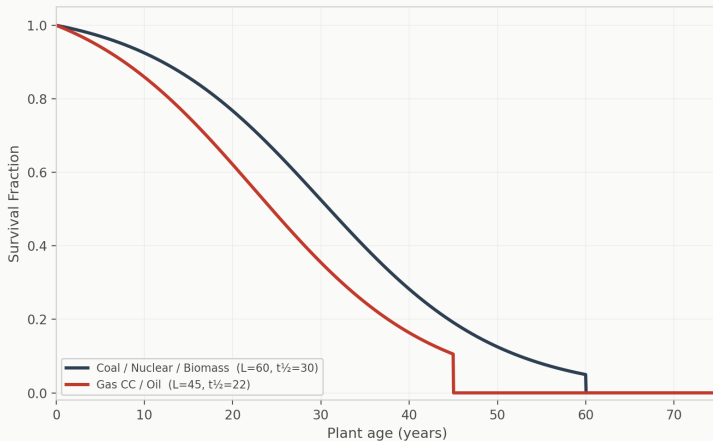
Profit-Based Shutdown $P(\pi)$

$$\pi = (p_{\text{elec}} - c_{\text{var}}) / |c_{\text{var}}|$$

- Smooth S-curve in profitability (median = -0.1)
- Carbon price $\uparrow \rightarrow \pi \downarrow \rightarrow$ coal exit

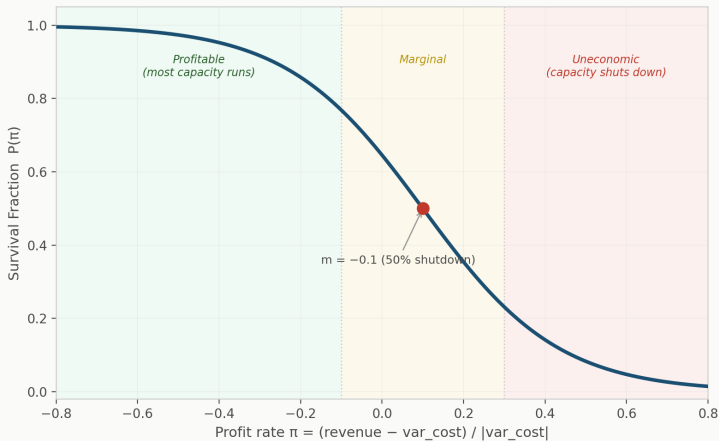
Vintage Retirement (Grubler, 1990)

- Age-based physical degradation — same-cohort plants retire at different ages



Profit Shutdown (Calvin et al., 2019)

- Economics-based early retirement — unprofitable plants shut down regardless of age



Results & Next Steps

Calibration Validation

Validate GHIM against GCAM v8.2 SSP2-Ref (32 regions, 2025–2100):

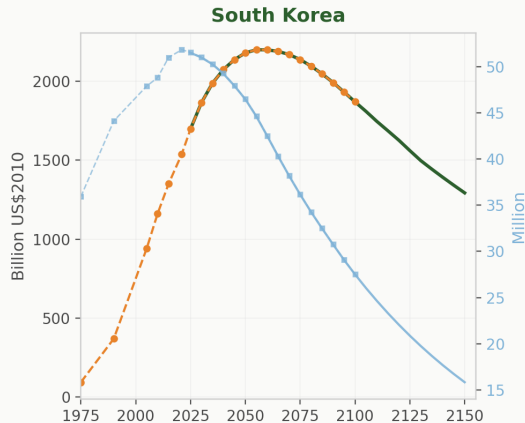
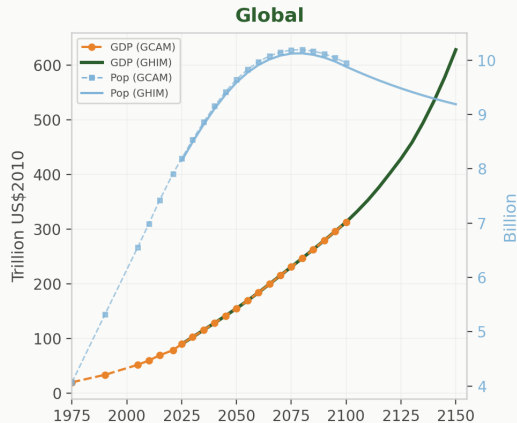
- GDP & Population
- Final Energy by Sector (Industry, Buildings, Transport)
- Final Energy by Carrier (Electricity, Gas, Liquids, Coal, Biomass, H₂, Heat)
- Electricity Generation
- Electricity Technology Shares

Calibration Statistics

- **CC** (Correlation Coefficient): trajectory shape agreement (−1 to 1, higher better)
- **MAPE** (Mean Absolute % Error): average relative deviation per period
- **MAE** (Mean Absolute Error): average absolute deviation (EJ or %p)

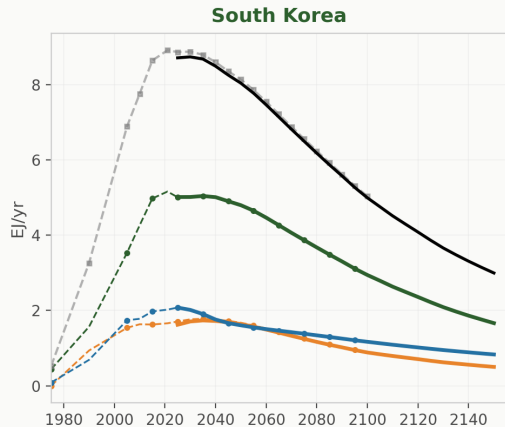
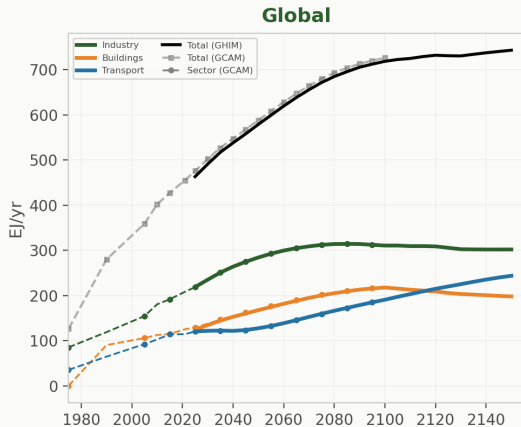
Calibration — GDP & Population

- **GDP:** $CC = 1.0000$, $MAPE = 0.01\%$, $MAE = 15.1 \text{ B\$}$
- **Population:** exogenous SSP2 projections (not calibrated)



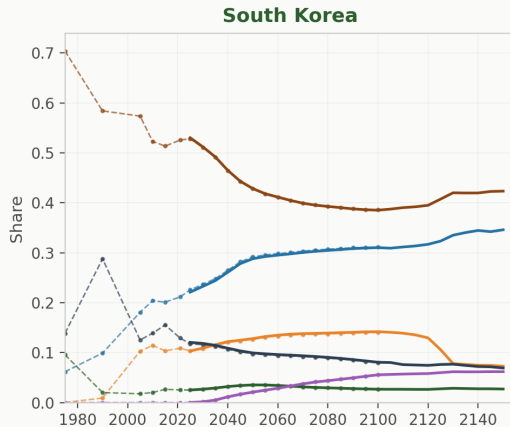
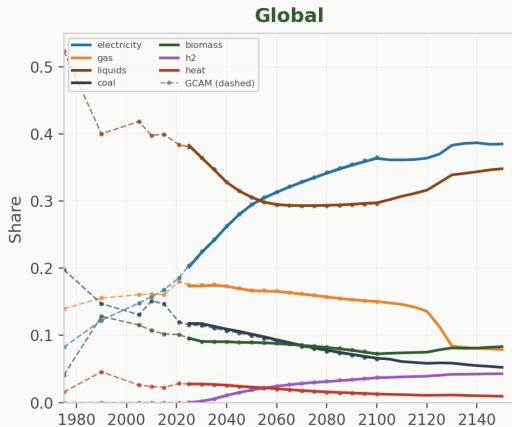
Calibration — Final Energy by Sector

- **Total Final Energy:** CC = 1.0000, MAPE = 1.5%, MAE = 9.1 EJ



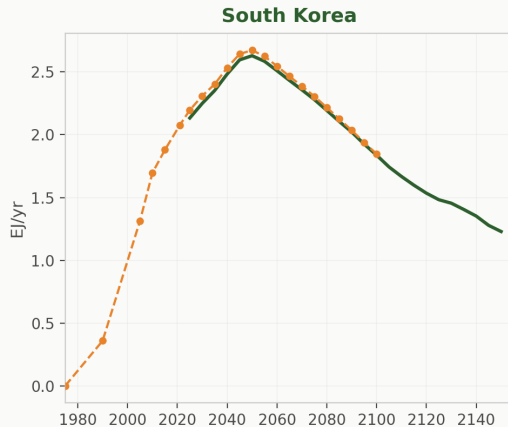
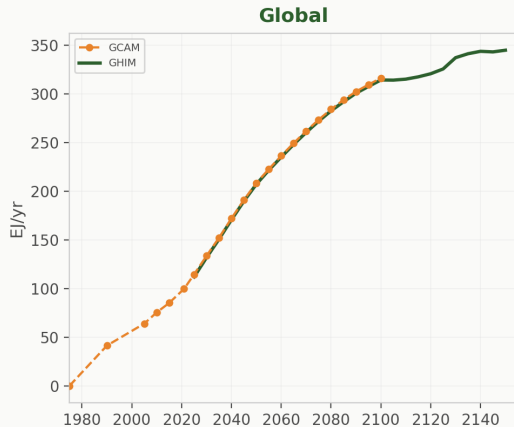
Calibration — Final Energy by Carrier

- Carrier Shares (share-weighted average): $CC = 0.9996$, $MAE = 0.2\%p$



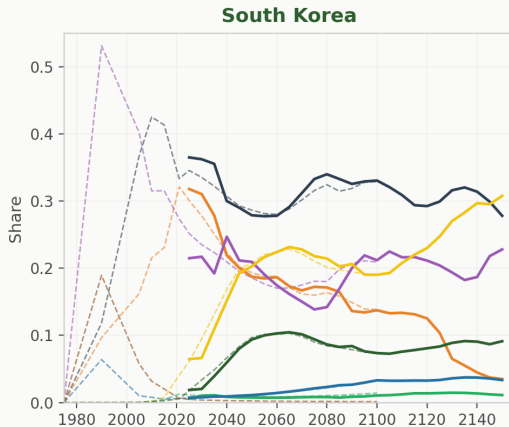
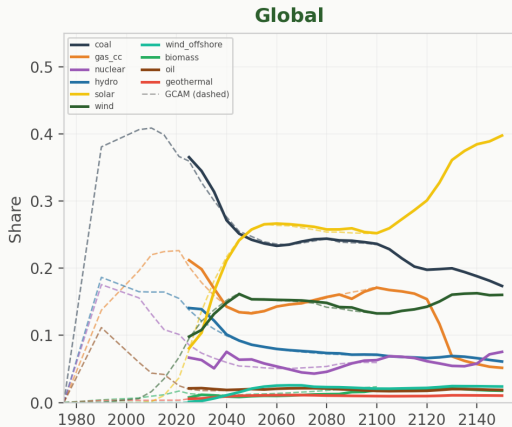
Calibration — Electricity Generation

- Electricity Generation: CC = 1.0000, MAPE = 0.8%, MAE = 1.6 EJ



Calibration — Electricity Generation Mix

- Technology Shares (generation-weighted average): $CC = 0.9450$, $MAE = 0.4\%$



Next Steps

■ 설계
 ■ 개발
 ■ 통합
 ■ 검증

작업	2026		2027		2028		2029		2030	
	상반기	하반기	상반기	하반기	상반기	하반기	상반기	하반기	상반기	하반기
1단계	1단계 — 2026~2027									
● 모형 설계	~2026.08 · 12개월									
● 모델 1.0 구축		~2027.08 · 18개월								
● 모델 1.0 커플링				Oneway with Land	~2028.06 · 12개월					
● 모델 1.0 검증					~2028.06 · 6개월					
2단계					2단계 — 2028~2030					
● 모델 2.0 설계					~2028.08 · 12개월					
● 모델 2.0 구축						~2029.08 · 18개월				
● 모델 2.0 커플링							Two way with Climate / Land / Water		~2030.08 · 18개월	
● 모델 2.0 검증									~2030.12 · 9개월	

Thank You

Questions & Discussion

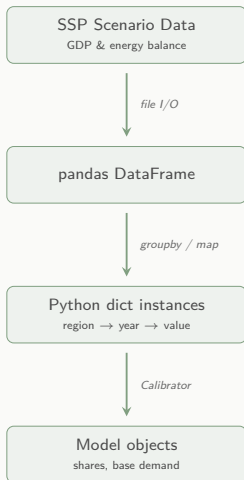
Appendix

Motivation: What GHIM Brings (Detailed)

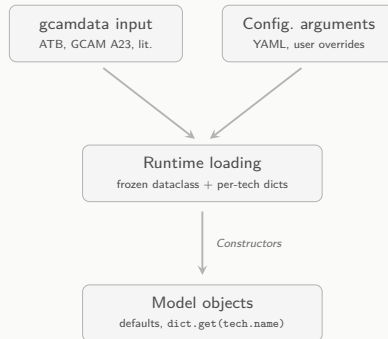
	Before	GHIM
Economy	· More energy detail, less macro feedback · Coupled models often soft-linked	· Energy $\leftrightarrow$ production feedback (KLEM) · Energy investment crowds out capital · Consistent capital, energy, materials accounting
Climate	· Soft coupling or offline · Mostly through damage function · Often one-sided	· Same-period hard coupling · Climate $\rightarrow$ Economy: multiple channels – TFP damage – Physical: efficiency, resources, demand · Energy $\rightarrow$ Climate via emissions
Technology	· Mostly fixed cost assumptions · Exception: WITCH (optimized RND)	· Two-factor learning (RND + LBD) · Global spillovers across technology families · Energy investment reduces available capital
Foresight	Myopic or perfect foresight	Adjustable rolling horizon
Resolution & Nesting	· ~32 regions, 5-year timesteps · Theoretically flexible · Hard to customize in practice	· 100+ countries, annual resolution · User-configurable sector nesting
Reporting	· Post-hoc translation for IPCC · Custom reporting possible but rarely used	· Direct AR6 IAMC-format output · Flexible reporting with user-defined functions

Data Pipeline: From Sources to Model Objects

Channel 1: Calibration Targets

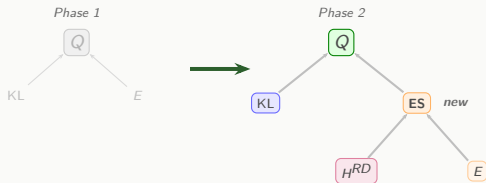


Channel 2: Structural Parameters



Phase 2 — Energy Services Nest (Popp, 2004)

Phase 1 → Phase 2



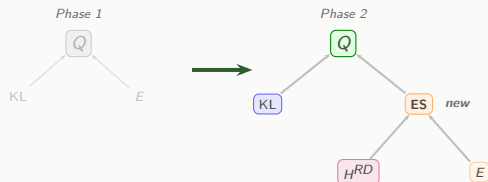
- $I_{RD}^{agg} = s_{RD} \cdot Y$ (GDP fraction)
- $I_{RD} \rightarrow H^{RD} \uparrow$ (knowledge stock)
- $\sigma_{ES} > 1$: H^{RD} substitutes for E
- E : EL + NEL (same as Phase 1)
- $ES = CES(H^{RD}, E; \sigma_{ES} > 1)$

What is aggregate efficiency?

- Cross-cutting R&D: insulation, smart grid, ...
- Same output, less energy $\rightarrow$ endogenizes AEEI

Phase 2 — Energy Services Nest (Popp, 2004)

Phase 1 → Phase 2



- $I_{RD}^{agg} = s_{RD} \cdot Y$ (GDP fraction)
- $I_{RD} \rightarrow H^{RD} \uparrow$ (knowledge stock)
- $\sigma_{ES} > 1$: H^{RD} substitutes for E
- E : EL + NEL (same as Phase 1)
- $ES = CES(H^{RD}, E; \sigma_{ES} > 1)$

What is aggregate efficiency?

- Cross-cutting R&D: insulation, smart grid, ...
- Same output, less energy → endogenizes AEEI

Energy Service Wraps Physical Energy

Phase 1: $E = ES$ (physical energy = energy service)

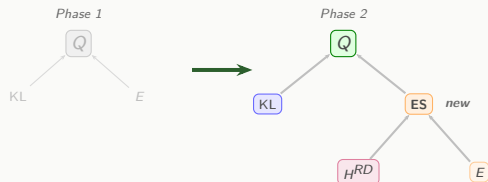
- E : physical energy input ($EJ \times$ base price)
- H^{RD} : knowledge stock (R&D-driven efficiency)
- $\sigma_{ES} > 1$: H^{RD} substitutes for E

→ Same **energy services**, less **physical energy**

→ Endogenous energy efficiency improvement

Phase 2 — Energy Services Nest (Popp, 2004)

Phase 1 → Phase 2



- $I_{RD}^{agg} = s_{RD} \cdot Y$ (GDP fraction)
- $I_{RD} \rightarrow H^{RD} \uparrow$ (knowledge stock)
- $\sigma_{ES} > 1$: H^{RD} substitutes for E
- E : EL + NEL (same as Phase 1)
- $ES = CES(H^{RD}, E; \sigma_{ES} > 1)$

What is aggregate efficiency?

- Cross-cutting R&D: insulation, smart grid, ...
- Same output, less energy → endogenizes AEEI

Energy Service Wraps Physical Energy

Phase 1: $E = ES$ (physical energy = energy service)

- E : physical energy input ($EJ \times$ base price)
- H^{RD} : knowledge stock (R&D-driven efficiency)
- $\sigma_{ES} > 1$: H^{RD} substitutes for E

→ Same **energy services**, less **physical energy**

→ Endogenous energy efficiency improvement

Unified Budget Constraint

$$s \cdot Y = I_K + I_{RD}^{agg} + \sum_j I_{RD,j}$$

- More R&D → less K short-term
but $H^{RD} \uparrow \rightarrow E \downarrow$ long-term

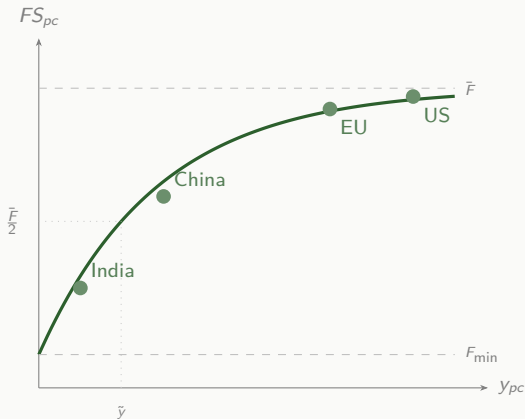
Final Energy Demand for Buildings (Eom et al., 2012) — 1/2: Floorspace

Step 1: Per-capita floorspace

Saturates with income — rich countries plateau.

$$FS_{pc} = (\bar{F} - F_{\min}) \left(1 - e^{\frac{-\ln 2}{\tilde{y}} y_{pc} (P/P_0)^\beta} \right) + F_{\min}$$

- $\bar{F}$: satiation level (m²/cap)
- $\tilde{y}$: midpoint income (50% of $\bar{F}$)
- $F_{\min}$: subsistence floor
- P/P_0 : housing price index (relative to base year)
- β : price elasticity of floorspace
- Floor: $FS_{pc} \geq FS_{pc, \text{base}}$ (no shrinkage)



Step 2: Total floorspace

$$FS(t) = FS_{pc}(t) \times \text{Pop}(t)$$

Final Energy Demand for Buildings (Eom et al., 2012) — 2/2: Service Intensity

Step 3: Service intensity per m²

Thermal load × service density per unit floor.

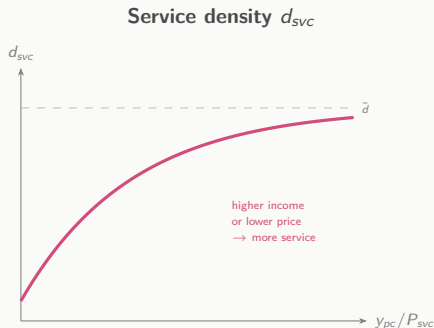
$$I_{svc}(t) = \underbrace{DD(t) \cdot U(t)}_{\text{thermal load}} \times \underbrace{d_{svc} \left(\frac{y_{pc}}{P_{svc}} \right)}_{\text{service density}}$$

- DD : degree days (HDD for heating, CDD for cooling)
- $U(t)$: shell conductance — improves over time
- d_{svc} : satiation function of *affordability* (y_{pc}/P_{svc})
- Non-thermal (lighting, etc.): no DD term

Step 4: Energy demand by service

$$E_{svc}(t) = FS(t) \cdot I_{svc}(t) \cdot \sum_c \frac{s_c}{eff_c^{svc}}$$

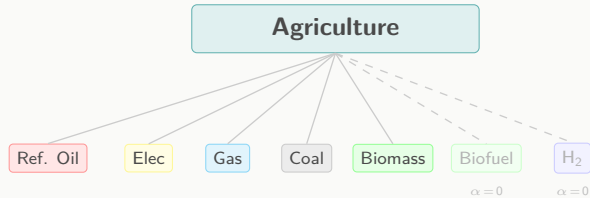
- s_c : carrier share from logit
- eff_c^{svc} : service efficiency of carrier c



$$P_{svc} = \sum_c s_c \cdot C_c / eff_c^{svc} \quad (\text{cost/GJ-service})$$

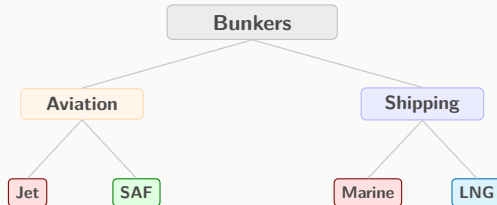
- Gas: $eff_{gas}^{svc} \approx 0.9$
- Elec: $eff_{elec}^{svc} \approx 3.0$ (heat pump)

Agriculture



- **Ag. Output:** $Q_{ag} = Q_0 \cdot \left(\frac{GDP}{GDP_0} \right)^{\alpha_{ag}} \cdot \left(\frac{P_{ag}}{P_0} \right)^{\gamma_{ag}}$
- **Energy:** $E_{ag} = ei_{ag} \times Q_{ag}$

Bunkers (International Aviation & Shipping)



- **Aviation:** $E_{avi} = E_0 \cdot (GDP_{world} / GDP_0)^{\alpha_{avi}}$
- **Shipping:** $E_{ship} = E_0 \cdot (GDP_{world} / GDP_0)^{\alpha_{ship}}$
- **LCOT:** $\frac{capex \cdot FCR}{annual-SD} + \frac{C_{fuel}}{eff} + O\&M$

Solver Design

Approach

1. Recursive-dynamic

Solve one period at a time (2025–2150)

$$\mathbf{x}_{n+1} = \alpha F(\mathbf{x}_n) + (1-\alpha) \mathbf{x}_n$$

2. State vector — 133 Dimensions

Regional ($\times 32$): ($Y_r, I_{E,r}, p_{elec,r}, p_{H_2,r}$)

Global ($\times 5$): $\mathbf{p}^W = (p_{coal}, p_{oil}, p_{gas}, p_{bio}, p_U)$

3. Implementation

$\alpha = 0.3$ (adaptive)

$\text{tol} = 5 \times 10^{-3}$, $\text{max_iter} = 150$

Inline pref recalibration per $F(\mathbf{x})$ call

Inter-period demand carry-forward (α_{ind})

Performance

Phase	Time
Data loading (Parquet $\rightarrow$ pandas)	0.9s
TFP calibration	36s
4 passes $\times$ 26 periods	
Forward solve	14s
26 periods $\times$ $\sim 0.5\text{s/period}$	
2 FE recal rounds + inline recal	
per iter: $F(\mathbf{x})$ over 32 regions	
Total	$\sim 50\text{s}$

Single-thread Python. **26/26 periods converge.**

Solver Iteration $F(\mathbf{x})$

